



Research paper

Model predictive attitude control of unmanned surface vehicle based on short-time wave prediction

Liang Hong^a, Haitao Liu^{a,b,*}, Quanshun Yang^a, Jiaxuan Yao^a

^a National Key Laboratory of Electromagnetic Energy, Naval University of Engineering, Wuhan, 430033, Hubei, PR China

^b East Lake Laboratory, Wuhan, 430204, Hubei, PR China

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ABSTRACT

Unmanned Surface Vehicles (USVs) are widely used in commercial and marine scientific research. In high sea conditions, USV is susceptible to significant random wave disturbances, necessitating the adoption of attitude control mechanisms to ensure safe operation. Existing control algorithms, mostly relying on Proportional Integral and Differential (PID) control and other common methods, tend to concentrate solely on the current model state, and seldom predict the future evolution of the model in combination with the state of the waves. This limitation hinders the optimization of attitude control in rough seas. To address this challenge, this paper proposes a model predictive attitude control approach grounded in Long Short-Term Memory neural network (LSTM) wave forecasting. By forecasting the wave disturbances on the USV in the coming period of time horizon and optimizing maneuvering commands, it identifies the optimal thruster actuation combination. This strategy dynamically adapts to sea conditions in real-time, enhancing the precision and accuracy of attitude control. The stability of this control strategy is validated using the discrete Lyapunov stability criterion. Simulation results demonstrate that our proposed control strategy exhibits remarkable roll and pitch reduction performance, surpassing PID control, linear quadratic regulator (LQR) control, model predictive control (MPC) without disturbance compensation, and MPC with disturbance compensation. Moreover, it maintains a superior control effect even in the presence of state feedback noise, indicating promising application prospects.

1. Introduction

Nowadays, Uncrewed Surface Vehicles (USVs) find widespread application across various sectors including commerce (Barrera et al., 2021), marine scientific research (Specht et al., 2017), and maritime stabilization operations (Patterson et al., 2022). USVs offer the advantages of affordability (Liu et al., 2016), autonomous navigation, and flexible maneuverability. They are equipped to carry instruments such as sonar, hydrographic detectors, and meteorological sensors, facilitating extensive and prolonged marine environment observations. Additionally, they are capable of executing military operations such as maritime patrol, reconnaissance, and underwater threat surveillance. However, these operations often encounter the challenge of adverse sea conditions. In high sea states, USVs contend with violent stochastic wave disturbances, leading to compromised stability and pronounced swaying motions, thereby impeding mission execution and posing safety risks. Hence, there arises a crucial need for attitude control techniques (including active roll, pitch alleviation and course keeping) to mitigate operational hazards associated with USVs.

At present, numerous scholars have conducted research on the hotpot of attitude control. These studies primarily fall into three main

categories: direct control, compensatory control, and predictive & compensatory control.

Direct control applies some generalized control algorithms to USV attitude stabilization. For example, PID control strategies (Surendran et al., 2007; Zhigang et al., 2011) are easy to implement in engineering, robust, and require low model accuracy. The PID control strategy is straightforward to implement in engineering, offering good robustness and low model accuracy requirements. However, when the USV operates in real-world scenarios, it encounters challenges stemming from the model's nonlinearity and random waves. Consequently, it becomes challenging for the PID controller to achieve rapid response, high accuracy, and a degree of self-adaptability for effective attitude control. To tackle the non-linear nature of the attitude control challenge, researchers Carletti et al. (2011), Lee et al. (2020), and Zhigang et al. (2018) have employed an improved sliding mode controller in their studies. Liu (2022), on the other hand, used the backstepping method to recursively perform the attitude control design with the assist of the Lyapunov stability criterion. For the uncertainty and randomness of the system, Pan et al. (2011) and Yang et al. (2002) then use robust control,

* Corresponding author at: National Key Laboratory of Electromagnetic Energy, Naval University of Engineering, Wuhan, 430033, Hubei, PR China.
E-mail address: liuht11@163.com (H. Liu).

which can ensure that the system can maintain its performance under the conditions of uncertainty factors, which is suitable for the uncertainty characteristics of wave disturbances. In order to further improve the control accuracy, a part of scholars proposed the control strategy based on neural networks, such as the neural network adaptive PD controller of Fang et al. (2012), and the method of Multi-dimensional Taylor Network (MTN) proposed by Zhang et al. (2017).

The above methods either utilize the robustness of the control algorithm at the expense of accuracy (e.g., PID, sliding mode control, robust control), or rely heavily on the accuracy of the model at the expense of some robustness (e.g., backstepping, neural network methods). In fact, in most cases, the mathematical model of USV is not completely accurate. In order to further improve the control accuracy and to ensure the robustness of the system, a series of compensating control strategies have been proposed in many literatures. Do and Pan (2001) used back propagation neural network to compensate the unknown nonlinear part of the navigator system. Li et al. (2016) proposed a radial basis function neural network to estimate the unmodeled nonlinear part of the system in order to solve the problem of output limitation and unknown parameters in the attitude control system. Luo et al. (2017), on the other hand, used a feed-forward neural network to take into account the uncertainty of the system, and the system performance outperformed the traditional PD controller.

While these control strategies are effective in mitigating model inaccuracy, they lack the capability to predict the model's future evolution, restricting their control to the current state. In contrast, Model Predictive Control (MPC) can forecast the motion state at a future time, enabling the implementation of high-performance attitude control.

Currently, the MPC attitude control strategy has been studied by many scholars (Fernandez and Hollinger, 2017; Jimoh et al., 2021; Li and Sun, 2011; Li et al., 2009; Liang and Wen, 2017; Liu et al., 2020). By observing the part of the USV's actual state that deviates from its mathematical model, i.e., "disturbance", and combining it with MPC, predictive & compensated attitude control can be realized. Li and Sun (2011) proposed the disturbance-compensated MPC control. The literature considers that the disturbance suffered by the navigator on the sea surface changes slowly, so the size of the disturbance measured at the previous moment is used as the disturbance value at the current moment to find the optimal compensation amount. Then the classical linear MPC optimization problem without considering the disturbance is solved to obtain the ideal control quantity, and finally the optimal compensation quantity is combined with the ideal control quantity to obtain the output of the final controller. Liang and Wen (2017) used a similar approach by designing a hysteresis observer to observe the disturbance at the previous moment and assuming that the perturbation is constant to the observer's observation during the MPC control period, and finally solving for the control quantity. In Jimoh et al. (2021), the state quantities and perturbations of the model are observed by designing optimal H_2 observers. The literature deforms the traditional MPC model by introducing control input increments as well as simulated disturbance inputs, and then the above two types of variables are used as the parameters to be optimized for the MPC to obtain the optimal input increments, from which the output of the final controller is then calculated.

However, none of the above methods really do the prediction of wave disturbance. They just roughly assume that the wave disturbances are constant within the prediction window of the model prediction, so there is still room for further improvement of the performance of the controller. To address this problem, this paper proposes an MPC control strategy based on LSTM short-term wave prediction for roll and pitch reduction and heading control of surface vehicles. MPC has an ability to look forward to the future of the model, and the predictive properties of LSTM can precisely strengthen this outlook ability, so it can effectively improve the control performance of the controller. Moreover, once the LSTM is trained, it does not need to occupy arithmetic resources, and the prediction effect is satisfying in general, which is suitable for the attitude stabilization of USVs.

The contributions of our article are listed below:

1. Models the USV in waves based on the Pierson–Moskowitz spectrum and linear wave theory;
2. Utilizes LSTM neural networks for short-term forecasting of wave disturbances;
3. Applies short-term wave forecast data to model predictive control to enhance the stability of the USV's attitude and demonstrates its stability;
4. Simulation experiments compare PID, LQR, MPC, and MPC with disturbance compensation control to the novel control strategy proposed in this paper, validating the superiority of our proposed control strategy under four different scenarios.

The rest of the paper is organized as follows. Section 2 delves into the mathematical model of the USV and the impact of wave force on it. Section 3 describes the conventional MPC control strategy, along with the proposed MPC+LSTM control strategy, and the stability of this new approach is also discussed. Section 4 presents simulations to validate the proposed control strategy and compares it against other control schemes, including PID, LQR, MPC without disturbance compensation, and MPC with disturbance compensation that uses current disturbance observations. Section 5 concludes the paper with a summary and the final conclusions regarding the control strategy.

2. Mathematical model

In order to validate the attitude control strategy proposed in this paper, the S175 model, which is publicly available, is chosen as the research object. The S175 has a similar mechanical structure and nonlinear characteristics as most USVs, and according to the simple principle of similarity, the performance of the control strategy proposed in this paper can be well examined.

For this purpose, a 6-DOF dynamics model of the USV as well as a wave model have to be established for simulating the attitude change of the USV under high sea state.

2.1. Dynamic model for surface vessel

Construct a 6-degree-of-freedom (6-DOF) nonlinear model of the S175 model as follow (Fossen, 2011):

$$(M_{RB} + M_A)\dot{v} + C_{RB}^*v + C_A^*v_r + Dv_r + \mu + G\eta = \tau_{wave} + \tau_{control} + \tau_{drag} \quad (1)$$

$$\begin{aligned} M_A &= A(\infty), C_A = U A(\infty)L \\ C_{RB} &= U M_{RB}L, D = B_{total}(\infty) \\ \mu &= \int_0^t K(t-\tau)(v(\tau) - Ue_1)d\tau \end{aligned} \quad (2)$$

In Eq. (1), M_{RB} is the inertia matrix of the system, including the mass matrix and inertia matrix of the system, M_A is the additional mass matrix, $C_{RB}(v)$ is the Coriolis matrix, C_A^* is the additional Coriolis matrix, D is the damping matrix, μ is the fluid memory effect vector, and G is the buoyancy matrix. For the sake of writing convenience, 'force' in this paper encompasses both force and moment. So τ_{wave} is the wave force, $\tau_{control}$ represents the control force and τ_{drag} denotes the drag force. In Eq. (2), $A(\infty)$ and $B_{total}(\infty)$ are the additional mass matrix and damping matrix at infinite frequency, and U is the bow velocity of the vehicle relative to the ocean current, which is the first component of the velocity vector. The matrix L is a 0–1 matrix and its value is:

$$L = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (3)$$

$v = [u \quad v \quad w \quad p \quad q \quad r]^T$ is the velocity vector of the navigator in the hull coordinate system.

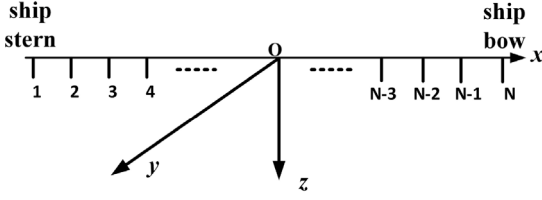


Fig. 1. Schematic diagram of drag force calculation by strip theory.

$v_c = [v_{xc} \ v_{yc} \ 0 \ 0 \ 0 \ 0]^T$ is the current velocity vector, $v_r = v - v_c$ is the relative velocity of the vehicle in the current, $\eta = [N \ E \ D \ \phi \ \theta \ \psi]^T$ is the actual position and attitude vector of the ship in the geodetic coordinate system. The relationship between v and η is satisfied as follows:

$$v = J(\eta)\dot{\eta}, J(\eta) = \begin{bmatrix} R & 0_{3 \times 3} \\ 0_{3 \times 3} & T \end{bmatrix} \quad (4)$$

$$R = \begin{bmatrix} \cos \psi \cos \theta & -\sin \psi \cos \phi + \cos \psi \sin \theta \sin \phi & \sin \psi \sin \phi + \cos \psi \cos \theta \sin \phi \\ \sin \psi \cos \theta & \cos \psi \cos \phi + \sin \psi \sin \theta \sin \phi & -\cos \psi \sin \phi + \sin \psi \cos \theta \sin \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (5)$$

$$T = \begin{bmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi / \cos \theta & \cos \phi / \cos \theta \end{bmatrix} \quad (6)$$

The 6-DOF mathematical model of S175 can be constructed by Eqs. (1) and (2). Also to be determined are the fluid memory effect μ , and the drag effect τ_{drag} , the wave force τ_{wave} . For the fluid memory effect μ , Fossen (2011) equated it to a state space model:

$$\begin{cases} \dot{x} = A_r x + B_r(v - U e_1) \\ \mu = C_r x \end{cases} \quad (7)$$

For the drag force τ_{drag} , the calculation is done by using strip theory. The coordinate origin is set at the midpoint of the ship's length, and the length of the ship is divided into N-1 equal segments (from the stern to the bow, as shown in Fig. 1).

Then the drag effect on the USV can be described as:

$$\begin{cases} \tau_{drag,1} = -A_x C_x \rho_w u_r |u_r| \\ \tau_{drag,2} = -\frac{1}{2} \rho_w \frac{A_y}{L_{pp}} C_D \sum_{i=1}^{N-1} f_v(i, r_r, v_r) |f_v(i, r_r, v_r)| dx \\ \tau_{drag,6} = -\frac{1}{2} \rho_w \frac{A_y}{L_{pp}} C_D \sum_{i=1}^{N-1} [dx(i-1) - \frac{L_{pp}}{2}] f_v(i, r_r, v_r) \cdot |f_v(i, r_r, v_r)| dx \end{cases} \quad (8)$$

$$f_v(i, r_r, v_r) = [dx(i-1) - \frac{L_{pp}}{2}] r_r + v_r$$

where $dx = L_{pp}/(N-1)$, A_x represents the forward projected area of the submerged part of the vehicle, while A_y denotes the lateral projected area of the submerged portion. C_x and C_D are the drag coefficients, ρ_w is the seawater density, and L_{pp} is the ship's length.

2.2. Wave model and wave force

In general, when considering waves of a single frequency and disregarding the effect of random phases, the relative height of the waves can be described by assuming the USV's speed as V , the wave frequency as ω , and the wave encounter angle as β :

$$\eta = A \cos[\omega_e t - (kx \cos \beta + ky \sin \beta)] \quad (9)$$

where A represents the wave amplitude, $k = \frac{\omega^2}{g}$ for the wave number, ω_e denotes the encounter frequency as $\omega_e = \omega - \frac{\omega^2}{g} U \cos \beta$, g stands

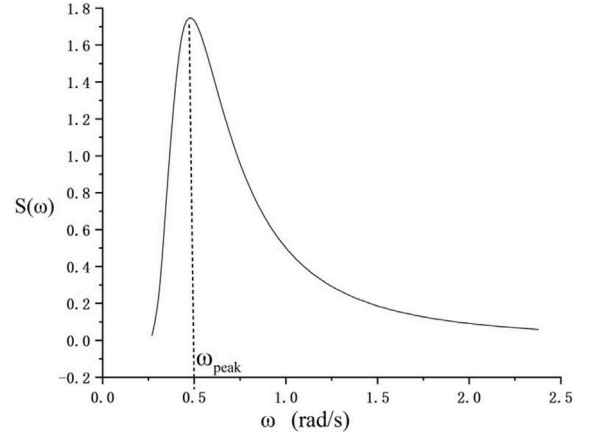


Fig. 2. Pierson-Moskowitz spectrum.

for gravitational acceleration, while x and y denote the horizontal and vertical coordinates of the vehicle's motion.

Actual ocean waves are complex and can be viewed as a superposition of waves with various frequencies. St Dinis et al. (1953) proposed using wave energy spectral functions to characterize waves in the frequency domain. In this paper, we employ the classical PM(Pierson-Moskowitz) wave spectrum for wave modeling.

$$S(\omega) = A\omega^{-5} \exp(-B\omega^{-4}) \quad (10)$$

where $A = 487H_s^2/T_0^4$ and $B = 1949/T_0^4$. H_s represents the significant wave height, T_0 denotes the modal period satisfying $T_0 = 2\pi/\omega_{peak}$, and ω_{peak} stands for the spectral peak frequency. Clearly, the energy integral of Eq. (10) yields $\int_0^\infty S(\omega)d\omega = \frac{A}{4B}$, and ω_{peak} can be estimated using the significant wave height according to ITTC (International Towing Tank Conference) standards:

$$\omega_{peak} = \frac{2\pi}{4.883 + 2.68H_s^{0.54}} \quad (11)$$

As shown above in Fig. 2, the PM spectrum is a single-peak spectrum, with the amplitude increasing from 0 to $\frac{5A}{4B\omega_{peak}} \exp(-5/4)$ as the frequency rises. After reaching the peak frequency, the amplitude decays rapidly.

However, the model described in Eq. (10) still does not adequately capture the time domain of a navigator's motion in waves. Longuet-Higgins (Longuet-Higgins and Deacon, 1957) thus proposed a linear wave theory model that accounts for long-crested waves and discretizes wave energy, which can be described as:

$$\eta = \sum_{i=1}^n \sqrt{2S(\omega_i)\Delta\omega} \cos[\omega_{ei}t + \varepsilon_i - (k_i x \cos \beta + k_i y \sin \beta)] \quad (12)$$

where n represents the number of wave frequencies, $S(\omega_i)$ is determined by Eq. (10), $\Delta\omega$ denotes the frequency's spatial length, with the maximum angular frequency for the simulation generally set between 3 and 5 times ω_{peak} . In this study, we set $5\omega_{peak}$ as the maximum angular frequency, hence $\Delta\omega = 5\omega_{peak}/n$. ω_i stands for a random angular frequency, where $\omega_i = (i-1)\Delta\omega + \kappa_i\Delta\omega$, and κ_i represents a uniformly distributed random number, with $\kappa_i \in [0, 1]$. k_i and ω_{ei} denote the wave number and encounter frequency at the i th wave frequency, respectively. ε_i signifies a random phase, satisfying $\varepsilon_i \in [0, 2\pi]$. During navigation on the water surface, changes in heading may occur, resulting in an encounter angle $\beta = \psi_{wave} - \psi$, defined as the difference between the wave direction angle and the heading angle.

According to the RAO (Response amplitude operator) data of S175, the wave force and moment subjected to the vehicle can be further

calculated as:

$$\tau_{wave,d} = \sum_{i=1}^n \{ A_{1,d} \cdot H_i \cos[\omega_{ei}t + \varepsilon_i - k_i(x \cos \beta_j + y \sin \beta_j) + P_{1,d}] + H_i^2 A_{2,d} \} \quad (13)$$

$$d \in \{1, 2, 3, 4, 5, 6\}, H_i = \sqrt{2S(\omega_i)\Delta\omega}$$

where $\tau_{wave,d}$ represents the wave force for a specific degree of freedom, $A_{1,d}$ and $P_{1,d}$ denote the increase and phase displacement of the first-order wave force for this degree of freedom, and $A_{2,d}$ represents the increase of the second-order wave force for this degree of freedom. The aforementioned amplifications and phase displacements are related to the ship's speed U , the wave encounter angle β , and the incoming flow angular frequency ω , which can be obtained by interpolating the RAO data set.

Obviously, Eq. (13) reveals that the first-order wave force is solely dependent on the wave amplitude and acts with the encounter frequency as a period. Conversely, the second-order wave action, proportional to the square of the wave amplitude, constitutes a non-periodic term. While the actual second-order wave action is considerably intricate, with certain components exhibiting periodicity (Choi et al., 2001), this paper simplifies it by focusing solely on the major non-periodic components and neglecting the minor ones.

3. Control method

This section summarizes the specific steps for the design of the model predictive attitude control strategy and analyzes the characteristics of the traditional control strategy within this framework. Optimization and improvement based on the limitations of the traditional control strategy are carried out, and the optimized model predictive attitude control method is formed, with its stability being proved.

3.1. Conventional MPC controller design

In general, MPC controller design can be divided into four steps:

Step1: Linearize the USV model to obtain a discrete state space model of the system. Configure the input as the control force affecting the 4th, 5th, and 6th degrees of freedom (roll, pitch and yaw), designate the output as the roll angle, pitch angle and yaw angle. Meanwhile, linearize the model considering a navigator speed of 5.1 m/s. Given that the fluid memory effect minimally impacts vehicle operation, this complexity is disregarded in the linearization process. Consequently, a continuous 9th-order state space model can be obtained:

$$\begin{cases} \dot{x} = A_c x + B_c u + B_w w \\ y = Cx \end{cases} \quad (14)$$

where, the model states is:

$x = [p \ q \ r \ \phi \ \psi \ \theta \ z \ v \ w]^T$, the input is: $u = [\tau_4 \ \tau_5 \ \tau_6]^T$, and the output is: $y = [\phi \ \theta \ \psi]^T$. $B_{w,c}$ represents the continuous input matrix of wave disturbances and the wave disturbance vector w satisfies $w = [w_K \ w_M \ w_N]^T$

The model above can be discretized with a sampling period T as follows:

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + B_w w_k \\ y_k = Cx_k \end{cases} \quad (15)$$

And the relationship between A_c , B_c and A , B satisfies:

$$\begin{cases} A = \exp(A_c T) \\ B = \int_0^T \exp(A_c \tau) B_c d\tau \\ B_w = \int_0^T \exp(A_c \tau) B_{w,c} d\tau \end{cases} \quad (16)$$

where $A : \mathbb{R}^n \times \mathbb{R}^n$ is the state matrix, $B : \mathbb{R}^n \times \mathbb{R}^p$ is the input matrix and $C : \mathbb{R}^m \times \mathbb{R}^n$ is the output matrix. u_k is the input vector, w_k is the wave disturbance input, and y_k is the output vector. In this paper, $n = 9, p = 3, m = 3$.

Step2: Define the cost function for the MPC attitude control problem. Denote the current state of the MPC as the k th step and define the cost function of the USV attitude control problem as:

$$J = \frac{1}{2} (y_{k+N_p} - y_{d,k+N_p})^T S (y_{k+N_p} - y_{d,k+N_p}) + \frac{1}{2} \sum_{l=0}^{N_p-1} [(y_{k+l} - y_{d,k+l})^T Q (y_{k+l} - y_{d,k+l}) + u_{k+l}^T R u_{k+l}] \quad (17)$$

where S is the end cost weight matrix and Q is the run weight matrix. $y_{d,k}$ denotes the expected attitude angle at the k th step, and N_p is the size of the prediction window of the MPC. The first term in Eq. (17) is the end cost, the second term is the run cost, and the third term is the control volume cost. The control function of the above optimization problem has an integral part, which is not conducive to simplifying the analysis and fast calculation. Therefore it may be useful to define:

$$X_k \triangleq \begin{bmatrix} x_{k+1,k} \\ x_{k+2,k} \\ \vdots \\ x_{k+N_p,k} \end{bmatrix}_{(nN_p) \times 1} \quad (18)$$

$$U_k \triangleq \begin{bmatrix} u_{k,k} \\ u_{k+1,k} \\ \vdots \\ u_{k+N_p-1,k} \end{bmatrix}_{(pN_p) \times 1} \quad (19)$$

$$\hat{W}_k \triangleq \begin{bmatrix} \hat{w}_{k,k} \\ \hat{w}_{k+1,k} \\ \vdots \\ \hat{w}_{k+N_p-1,k} \end{bmatrix}_{(pN_p) \times 1} \quad (20)$$

and set:

$$X_k = \Phi x_{k,k} + \Gamma U_k \quad (21)$$

Then according to Eq. (17), it can be deduced that:

$$\Phi \triangleq \begin{bmatrix} A \\ A^2 \\ \vdots \\ A^{N_p} \end{bmatrix}_{(nN_p) \times n} \quad (22)$$

$$\Gamma = \begin{bmatrix} B & 0 & \dots & 0 \\ AB & B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N_p-1}B & A^{N_p-2}B & \dots & B \end{bmatrix}_{(nN_p) \times (pN_p)} \quad (23)$$

$$\Gamma_w = \begin{bmatrix} B_w & 0 & \dots & 0 \\ A B_w & B_w & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A^{N_p-1}B_w & A^{N_p-2}B_w & \dots & B_w \end{bmatrix}_{(mN_p) \times (nN_p)} \quad (24)$$

and for Eq. (22), 0 represents the zero matrix, and the following text applies likewise.

Noticed that, $x_{k,k}$ and $y_{d,k}$ are constant. So we can ignore the term that does not contain U_k . Through Eqs. (21) (22) (23) (24), Eq. (17) can be rewritten as:

$$J = U_k^T \Gamma^T C_0^T \Omega [C_0 \Phi x_{k,k} + C_0 \Gamma_w \hat{W}_k - y_{d,k}] + \frac{1}{2} U_k^T [\Gamma^T C_0^T \Omega C_0 \Gamma + \Psi] U_k \quad (25)$$

where:

$$C_0 = \begin{bmatrix} C & 0 & 0 & 0 \\ 0 & C & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & C \end{bmatrix}_{(mN_p) \times (nN_p)} \quad (26)$$

$$\Omega = \begin{bmatrix} Q & 0 & 0 & 0 \\ 0 & Q & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & S \end{bmatrix}_{(mN_p) \times (mN_p)} \quad (27)$$

$$\Psi = \begin{bmatrix} R & 0 & 0 & 0 \\ 0 & R & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R \end{bmatrix}_{(pN_p) \times (pN_p)} \quad (28)$$

$$Y_{d,k} = [y_{d,k+1} \quad y_{d,k+2} \quad \cdots \quad y_{d,N_p}]^T \quad (29)$$

Step3:Determine the constraints of the MPC optimization problem. In practical control scenarios, the controller outputs and state variables are subject to certain limitations, outlined as follows:

$$\begin{cases} u_{\min} \leq u_{k+i-1} \leq u_{\max} \\ \Delta u_{\min} \leq u_{k-1+i} - u_{k-2+i} \leq \Delta u_{\max} \\ x_{\min} \leq x_{k+i} \leq x_{\max} \\ y_{\min} < y_{k+i} < y_{\max} \end{cases} \quad (30)$$

$i = 1, 2, \dots, N_p$

The first term denotes a limit on the controller output, while the second term denotes a limit on the rate of change of the controller output, the third term denotes a limit on the amount of state at each moment, and the fourth term denotes a limit on the output.

After simplifying Eq. (30), we shall safely yield the following optimization problem:

$$U_k = \arg \min_{U_k} J$$

$$s.t. \begin{cases} u_{\min} \otimes H \leq U_k \leq u_{\max} \otimes H \\ H \otimes \Delta u_{\min} \leq S \begin{bmatrix} u_{k-1} \\ U_k \end{bmatrix} \leq H \otimes \Delta u_{\max} \\ H \otimes x_{\min} \leq \Phi x_{k|k} + \Gamma U_k \leq H \otimes x_{\max} \\ C_0 \cdot (H \otimes x_{\min}) \leq C_0 \cdot (\Phi x_{k|k} + \Gamma U_k) \\ \leq C_0 \cdot (H \otimes x_{\max}) \end{cases} \quad (31)$$

where:

$$H = [1 \quad 1 \quad \cdots \quad 1]_{1 \times N_p}^T$$

$$H_p = [-1 \quad 0 \quad 0 \quad \cdots \quad 0 \quad 1]_{1 \times (p+1)}$$

$$S = \begin{bmatrix} S_1 \\ S_2 \\ \vdots \\ S_i \\ \vdots \\ S_{p \times N_p} \end{bmatrix}, S_i = [0_{1 \times (i-1)} \quad H_p \quad 0_{1 \times (pN_p + N_p - p - i)}] \quad (32)$$

In Eq. (31), ‘ $\otimes$ ’ denotes the Kronecker product, H represents the full ‘1’ vector, and H_p is the row vector with -1 in the first position and 1 in the last position. Both u_{k-1} and $x_{k|k}$ are known, thus all constraints in Eq. (30) are transformed into constraints solely related to U_k .

Step4:Solve the MPC optimization problem. Eq. (31) represents a typical quadratic programming problem with constraints, making it solvable using MATLAB’s quadprog function. Upon obtaining U_k , the initial component is selected as the control input at step k . Subsequently, U_{k+1} is computed, and this process continues, continuously advancing the window to achieve MPC attitude control.

3.2. Analysis of traditional model prediction attitude control methods

There are three typical modes of traditional MPC active shake reduction controller design. They are MPC attitude control (i.e., without disturbance compensation, later abbreviated as MPC), active attitude control based on feedforward compensation of a nominal system (Liang and Wen, 2017) (abbreviated as MPC+FC1), and feedback compensated attitude control based on the system with disturbances (Jimoh et al., 2021) (abbreviated as MPC+FC2).

The difference between the three typical MPC attitude control strategies is mainly reflected in Eq. (25):

- For the MPC method, the nominal system is generally used directly for model prediction, therefore, the disturbance term w_k in Eq. (15) is neglected, at which point $\hat{W}_k \equiv 0$. Without loss of generality, Eq. (25) degenerates by assuming $Y_{d,k} = 0$:

$$J = U_k^T \Gamma^T C_0^T \Omega [C_0 \Phi x_{k,k}] + \frac{1}{2} U_k^T [\Gamma^T C_0^T \Omega C_0 \Gamma + \Psi] U_k \quad (33)$$

- For MPC+FC1, the nominal system is also used for model prediction, however, it uses the disturbances as a feed-forward compensation term, i.e., let $y_{d,k+l} = w_{k-1}$, $l = 0, 1, 2, \dots, N_p - 1$ in $Y_{d,k}$, then Eq. (25) becomes:

$$J = U_k^T \Gamma^T C_0^T \Omega [C_0 \Phi x_{k,k} - Y_{d,k}] + \frac{1}{2} U_k^T [\Gamma^T C_0^T \Omega C_0 \Gamma + \Psi] U_k \quad (34)$$

where for the disturbance w_{k-1} , according to Eq. (15), if all states of the model are measurable, there we have:

$$w_{k-1} = C(x_k - Ax_{k-1} - Bu_{k-1}) \quad (35)$$

- For MPC+FC2, the prediction is made using a model with disturbance, i.e., keeping the cost function of Eq. (25) and observing w_{k-1} through Eq. (35), while making $\hat{w}_{k+l,k} = w_{k-1}$, $l = 0, 1, 2, \dots, N_p - 1$ in $\hat{W}_k$. In Jimoh et al. (2021), the model states are not directly available. So it design the H_2 optimal observer for the observation w_k , which is beyond the scope of this paper and will not be discussed here.

It can be seen that traditional MPC controllers do not reasonably predict the amount of the disturbance of the model within N_p steps, either assuming the disturbance $\hat{W}_k \equiv 0$, or directly using the observed disturbances as the disturbances of the backward N_p steps, which limits the control performance of the MPC. In the later paper, we will address this problem by introducing LSTM neural network to reasonably predict the disturbance and improve the control accuracy of MPC.

3.3. Improved design with LSTM network

As mentioned in the previous section, most of the articles on wave disturbance compensation do not “predict” wave disturbances, and most of them consider wave disturbances as slowly changing quantities, or only consider wave disturbances in the current state.

There are two schools of thought on the methods of wave prediction. The first employs a model-based prediction approach, exemplified by Booij et al. (1999), which utilizes models to simulate realistic waves. The second method relies on machine learning techniques, such as those explored by Deo and Naidu (1998). While these methods enable wave prediction, they provide only approximate forecasts and are not yet suitable for wave compensation, highlighting the necessity for short-term wave forecasting. Hao et al. (2022) and Zhao et al. (2023) preprocessed wave data and subsequently employed LSTM neural networks for short-term wave forecasting.

Model predictive compensatory control based on wave prediction has been applied in the fields of WECs (Wave Energy Converters) (Coe et al., 2018), power positioning (Øveraas et al., 2023). Among them, Øveraas et al. (2023) gives a model-based approach for short-term wave prediction using MPC for ship power positioning. However, compared to wave generator and power positioning, USV has a certain heading, speed and more degrees of freedom, which will increase the difficulty of wave forecasting. Simulations show that the smaller frequency and phase deviations of the model-based short-time wave forecast in this scenario can also lead to inaccurate prediction, while a larger arithmetic power is required to further improve the forecasting performance.

Based on the above, this section proposes to use LSTM to learn the pattern of wave disturbances and forecast the wave disturbances in the future period, and then use MPC for attitude control.

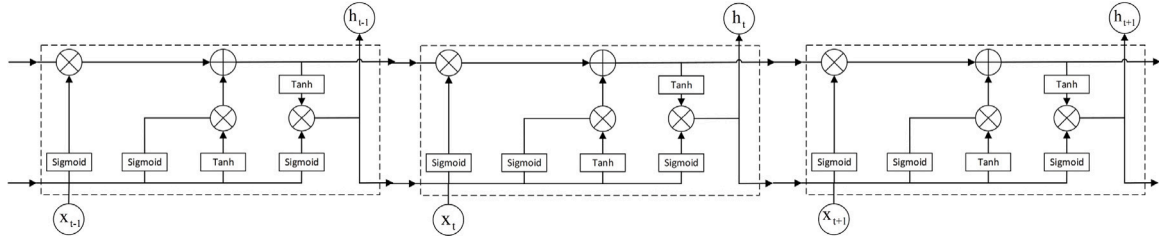


Fig. 3. LSTM neural network.

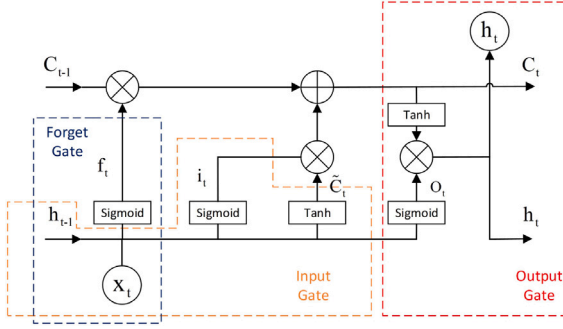


Fig. 4. Unit structure of LSTM neural networks.

By focusing on the disturbances that directly influence the observed attitude, wave disturbances can be monitored using Eq. (35), i.e. $w_{k-1} = C(x_k - Ax_{k-1} - Bu_{k-1})$. The physical meaning of Eq. (35) is that at time $k-1$, the system experiences a state quantity change of $w(k-1)$ at time k solely due to the wave force. This implies that MPC initiates each control step by assessing the wave disturbance from the preceding step. However, this poses a paradox as the wave action must have already occurred for the observer to calculate the disturbance magnitude. If the amount of the disturbance can be forecasted in short time based on the historical data of wave disturbance, it can be foreseen that this method will surely improve the control performance of the MPC attitude control.

LSTM (Long Short Term Memory), as shown in Fig. 3, is a unique form of recursive neural network. The commonly used feed-forward neural networks treat inputs at time t and time $t+1$ as unrelated, however, the actual wave action, according to Eq. (13), is obviously associated with the time dimension. LSTM neural network, effectively alleviate the traditional RNN “long-term dependence problem”, “gradient disappearance problem”.

As shown in Fig. 4, the LSTM neural network has three gates, i.e., forget gate, input gate, and output gate.

The first gate is the Oblivion Gate, which is used to select which information should be retained, and is updated by:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t]^T + b_f) \quad (36)$$

where σ is the Sigmoid function that satisfies $\sigma(x) = 1/(1 + e^{-x})$.

The second gate, known as the input gate, is responsible for determining the state update of the cell C_t , and its calculation is as follows:

$$\begin{aligned} C_t &= f_t * C_{t-1} + i_t * \tilde{C}_t \\ i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \end{aligned} \quad (37)$$

The third gate, referred to as the output gate, is crucial in determining the predicted value of the LSTM neural network, and its calculation is as follows:

$$h_t = \sigma(W_O \cdot [h_{t-1}, x_t] + b_O) * \tanh(C_t) \quad (38)$$

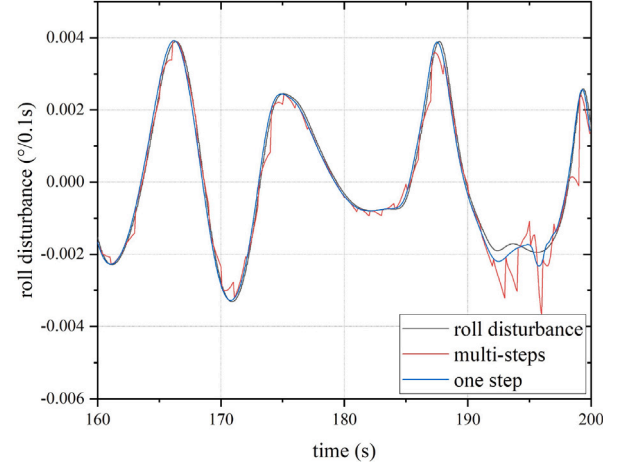


Fig. 5. Comparison of the effectiveness between single-step forecasting and multi-step forecasting (10 steps).

where “*” refers to the multiplication of corresponding elements in a matrix and “.” denotes matrix multiplication.

For the neural network parameters, this paper uses the Adam algorithm and error gradient backpropagation to continuously update the eight parameters ($W_f, W_i, W_C, W_O, b_f, b_i, b_C, b_O$) until convergence criteria are met.

Due to the characteristics of the LSTM neural network, wave forecasting can be categorized into single-step forecasting and multi-step forecasting. In the single-step forecasting model, the wave disturbance w_{k-1} observed by the current observer is known. The cell state of the LSTM neural network is then updated based on w_{k-1} , and the wave disturbance $\hat{w}_k$ at the next moment is forecasted accordingly.

The multi-step prediction model is different from the single-step (assume that the deduce steps is N_p), it knows w_{k-1} , and predicts $\hat{w}_k$ in the same way, but continues to use $\hat{w}_k$ to predict the wave disturbance $\hat{w}_{k+1}$ at the next moment until it predicts $\hat{w}_{k+N_p-1}$, then obtains a new disturbance observation w_{k+N_p-1} . This cycle is repeated to complete the wave disturbance prediction. Compared with single-step prediction, multi-step prediction has the problem of “error accumulation”. While multi-step prediction is essential for Model Predictive Control (MPC), it does introduce the challenge of “error accumulation” compared to single-step prediction. Therefore, mitigating the issue of error accumulation resulting from multi-step prediction is crucial (see Fig. 5).

In this paper, we consider the “parallel forecasting” scheme, where single-step forecasting and multistep forward-looking forecasting are carried out simultaneously. To elaborate, single-step forecasting takes place at time k to generate the single-step network $LSTM_1$. Concurrently, multistep forecasting is conducted to predict the wave disturbances $\hat{W}_k = [\hat{w}_k \ \hat{w}_{k+1} \ \dots \ \hat{w}_{k+N_p-1}]$ for N_p time steps, which is then integrated into the model prediction. At this point, the network $LSTM_2$ accumulates errors from the multistep forecast. Consequently, $LSTM_1$ is substituted for $LSTM_2$ in the multistep forecast network for

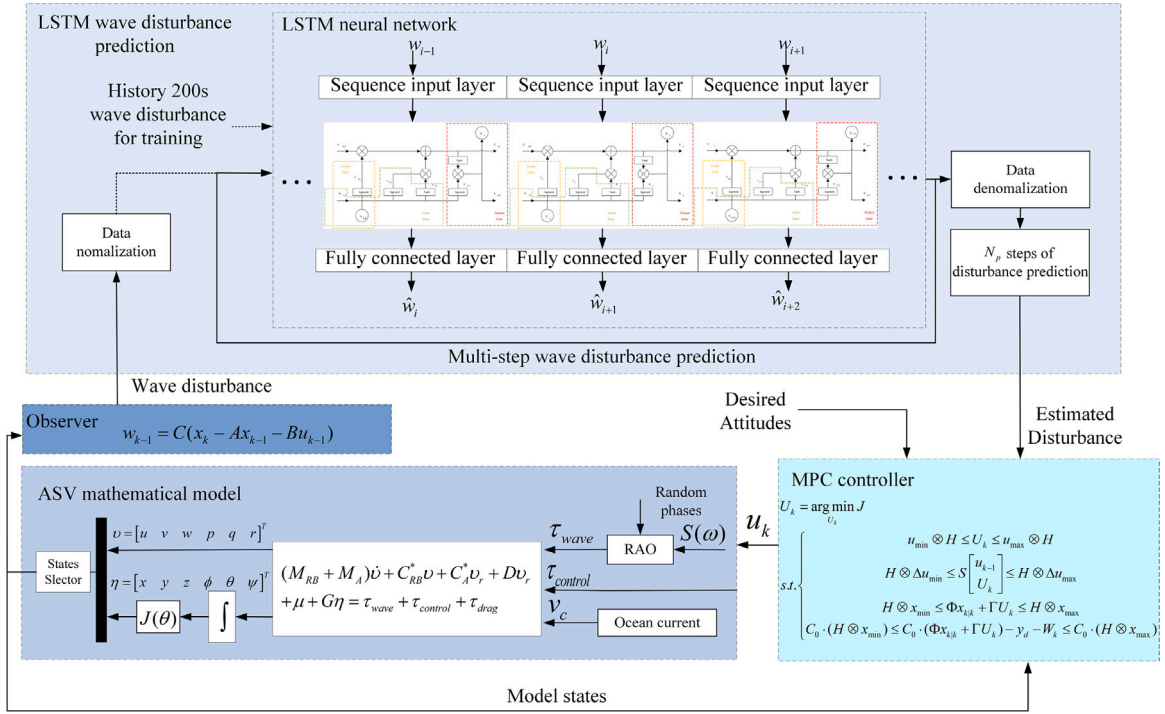


Fig. 6. Wave compensated MPC attitude control strategy based on LSTM.

the $k+1$ th step of wave disturbance prediction. This refined forecasting approach effectively mitigates the error accumulation issue associated with multistep forecasting, thus enhancing forecasting accuracy.

The control strategy used in this paper is shown in Fig. 6. First, the LSTM neural network needs to undergo pre-training. The historical disturbance data from the past 200 s, including roll, pitch and yaw data, are chosen as the training set. This set is normalized before being fed into the LSTM neural network through a sequential input layer for training. Unlike traditional LSTM networks, the output of this network will be linked to a fully-connected layer to produce cell states based on their respective weights.

At the k th time step, the wave disturbance observer computes the disturbance w_{k-1} , based on the current state of the USV. Subsequently, this disturbance data is normalized and forwarded to the LSTM neural network for ‘parallel prediction’ and then denormalized to acquire wave disturbances of N_p time steps. These disturbances are utilized for model prediction within the MPC framework. The MPC is then optimized to determine the control inputs, enabling the attitude control of the USV.

The disturbance estimation matrix in Eq. (20), then, can be replaced by the short-time forecast value of LSTM, i.e., make the forecast of the LSTM multistep forecast network at step k for the wave disturbance at step i as $\hat{w}_{k+i,k}$.

At this point, the optimization problem to be solved is shown in Eq. (39), which is a typical quadratic programming problem with constraints:

$$\begin{aligned}
 U_k &= \arg \min_{U_k} J, \\
 J &= U_k^T \Gamma^T C_0^T \Omega [C_0 \Phi x_{k,k} + C_0 \Gamma_w \hat{W}_k - Y_{d,k}] + \\
 &\quad \frac{1}{2} U_k^T [\Gamma^T C_0^T \Omega C_0 \Gamma + \Psi] U_k \\
 \text{s.t.} \quad &\begin{cases} u_{\min} \otimes H \leq U_k \leq u_{\max} \otimes H \\ H \otimes \Delta u_{\min} \leq S[U_k] \leq H \otimes \Delta u_{\max} \\ H \otimes x_{\min} \leq \Phi x_{k,k} + \Gamma U_k \leq H \otimes x_{\max} \\ C_0 \cdot (H \otimes x_{\min}) \leq C_0 \cdot (\Phi x_{k,k} + \Gamma U_k + \Gamma_w \hat{W}_k) \\ \leq C_0 \cdot (H \otimes x_{\max}) \end{cases} \quad (39)
 \end{aligned}$$

3.4. Stability proof

To demonstrate the stability of the novel control strategy in Section 3.2, the navigator model can be expressed by considering the LSTM prediction error δw_k :

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + B_w \hat{w}_k + \delta w_k \\ y_k = Cx_k \end{cases} \quad (40)$$

We can view $\hat{w}_k$ as a component of the model that remains known at any given moment (predictable by the LSTM), while δw_k represents an error disturbance that eludes prediction by the LSTM. Thus, stability of the control strategy hinges on identifying a feasible domain and a stabilizable control rate. Let $e_k = x_k - \bar{x}_k$, and decompose the system into two parts: the nominal system and the error system, as follows:

$$\begin{cases} \bar{x}_{k+1} = A\bar{x}_k + B\bar{u}_k + B_w \hat{w}_k \\ e_{k+1} = Ae_k + \delta w_k \end{cases} \quad (41)$$

Therefore, the control strategy proposed in this paper is considered stable if the stability of the nominal system can be proven and if the bounded value e_k does not cause the state quantities to exit the stability domain.

To prove the stability of the system, let us make the following reasonable assumptions:

Assumption 1. The size of the wave disturbance predicted by the LSTM is finite and satisfies: $\hat{w}_k \in \Theta_w, 0 < k$;

Assumption 2. The error in LSTM wave forecasting is finite and satisfies: $\delta w_k = w_k - \hat{w}_k \in \Theta_{\delta w}, 0 < k$;

Assumption 3. Without loss of generality, let the desired output be $y_{d,k} = [0 \ 0 \ 0]^T, 0 < k$.

Assumption 4. In Eq. (15), (A, B_u) can be stabilized. We disregard the second and third constraints of Eq. (31), meaning there are no restrictions on the magnitude of the input rate of change. Thus, the

constraint of the optimization problem becomes:

$$s.t. \begin{cases} u_{\min} < u < u_{\max}, \forall x \in \Theta \\ y_{\min} < y < y_{\max} \end{cases} \quad (42)$$

3.4.1. Stability of nominal system

In order to prove the stability of the nominal system in Eq. (42), some necessary preparations are made first (Rawlings et al., 2017):

Definition 1. $A \oplus B := \{a+b | a \in A, b \in B\}$; $A \dot{-} B := \{x \in \mathbb{R}^n | \{x\} \oplus B \subseteq A\}$; $KA := \{Ka | a \in A\}$, $K \in \mathbb{R}^{m \times n}$; $K \odot A := \{x \in \Theta_i | Kx \in A\}$, $K \in \mathbb{R}^{m \times n}$, the default is $x \in \Theta$ if not otherwise specified in the text.

Definition 2. If the continuous function $\alpha : [0, a] \rightarrow [0, \infty)$ is strictly increasing and $\alpha(0) = 0$, then α belongs to the class of $\mathcal{K}$ functions. If $a = \infty$, and $r \rightarrow \infty$, when $\alpha(r) \rightarrow \infty$, then α belongs to the $\mathcal{K}_\infty$ class of functions.

Lemma 1. If Assumption 4 holds, there must be a state feedback $u = Kx$ such that $A+BK$ is asymptotically stable, and there must be a neighborhood Θ of equilibrium points where the system satisfies the constraints of Eq. (42) with $(A+BK)x \in \Theta$. Proof omitted; Sontag (2009)

Lemma 2. For the system:

$$x(k+1) = f(x(k)), k \geq 0, x(0) = x_0 \quad (43)$$

where f is continuous with respect to x and $f(0) = 0$, then if there exists a positive definite bounded function $V(x)$ satisfying:

1. $V(0) = 0$, and $V(x) > 0$ for $x \neq 0$,
2. Near the equilibrium point $x = 0$, $V(x)$ is continuous,
3. Along the system trajectory, there has:

$$V(x(k+1)) - V(x(k)) \leq -\alpha_0(\|x(k)\|) \quad (44)$$

where $\alpha_0(\cdot)$ is a class $\mathcal{K}$ function,

then the system is asymptotically stable. The proof is omitted. Diehl et al. (2010)

Theorem 1. Suppose $P > 0$ is a positive definite solution of the Lyapunov equation

$$P = (A+BK)^T P (A+BK) + (C^T Q C + K^T R K) \quad (45)$$

then there exists $\alpha > 0$ such that in the neighborhood defined by (P, α) :

$$\Theta(P, \alpha) \doteq \{x \in \mathbb{R}^n | x^T P x \leq \alpha\} \quad (46)$$

system (15) satisfies the constraints of Eq. (42)

Proof of Theorem 1. By the definition of Θ , it follows that the element $0 \in \Theta$, so that $u_{\min} < 0, y_{\min} < 0, u_{\max} > 0, y_{\max} > 0$, from this, we can obtain:

$$((A+BK)x)^T P ((A+BK)x) = x^T (A+BK)^T P (A+BK)x$$

if $P > 0$ and Eq. (45) is satisfied, then we have:

$$\begin{aligned} & ((A+BK)x)^T P ((A+BK)x) \\ &= x^T P x - x^T (C^T Q C + K^T R K)x \\ &\leq x^T P x \end{aligned}$$

And since $x \in \Theta$, so $x^T P x \leq \alpha$, then we have:

$$((A+BK)x)^T P ((A+BK)x) \leq \alpha, \forall x \in \Theta$$

Therefore, Theorem 1 is proved.

Assuming N_p as the control window size for the model, Eq. (39) simplifies to:

$$\begin{aligned} & \min_{U_k} J, \\ & J = \frac{1}{2} \left[\sum_{i=0}^{N_p-1} (\bar{y}_{k+i,k}^T Q \bar{y}_{k+i,k} + u_{k+i,k}^T R u_{k+i,k}) \right] + \frac{1}{2} \bar{x}_{k+N_p}^T P \bar{x}_{k+N_p} \\ & s.t. \begin{cases} u_{\min} \leq u_{k+i,k} \leq u_{\max}, i = 0, 1, \dots, N_p - 1 \\ \bar{y}_{\min} \leq \bar{y}_{k+i,k} \leq \bar{y}_{\max}, i = 1, 2, \dots, N_p \\ \bar{x}_{k+N_p,k} \in \Theta \dot{-} (-CBK) \odot \Theta_w \end{cases} \end{aligned} \quad (47)$$

Theorem 2. For the nominal system in Eq. (42), suppose that at time $k = 0$, problem (48) has a solution and P satisfies Lyapunov's Eq. (45). Θ in problem (47) is defined by Eq. (46) and satisfies the constraints; y is zero-state observable. Then, for $\forall k > 0$, problem (47) is solved by updating the state quantity $x_{k,k}$, and the nominal system in the closed-loop system (41) becomes asymptotically stable.

Proof of Theorem 2. Assume that the optimization problem has a solution at moment k and denote it as:

$$U_k^* = \begin{bmatrix} u_{k,k}^* \\ u_{k+1,k}^* \\ \vdots \\ u_{k+N_p-1,k}^* \end{bmatrix}_{pN_p \times 1}$$

The corresponding predicted state sequence and predicted output sequence are denoted as:

$$\begin{cases} \{\bar{x}_{k+1,k}^*, \bar{x}_{k+2,k}^*, \dots, \bar{x}_{k+m,k}^*\} \\ \{\bar{y}_{k+1,k}^*, \bar{y}_{k+2,k}^*, \dots, \bar{y}_{k+m,k}^*\} \end{cases}$$

Therefore, The corresponding optimal cost function can be transformed into:

$$J_k^* = \frac{1}{2} \left[\sum_{i=0}^{N_p-1} (\bar{y}_{k+i,k}^{*T} Q \bar{y}_{k+i,k}^* + u_{k+i,k}^{*T} R u_{k+i,k}^*) \right] + \frac{1}{2} \bar{x}_{k+N_p}^{*T} P \bar{x}_{k+N_p}^* \quad (48)$$

By substituting $u_{k,k}^*$ into Eq. (41), we obtain the state of the system at time moment $k+1$:

$$\bar{x}_{k+1} = A \bar{x}_k + B u_{k,k}^* + B_w \hat{w}_k \quad (49)$$

Let us choose a control sequence at moment $k+1$:

$$U_{k+1} = \begin{bmatrix} u_{k+1,k+1} \\ u_{k+2,k+1} \\ \vdots \\ u_{k+N_p-1,k+1} \\ u_{k+N_p,k+1} \end{bmatrix}_{pN_p \times 1} \doteq \begin{bmatrix} u_{k+1,k}^* \\ u_{k+2,k}^* \\ \vdots \\ u_{k+N_p-1,k}^* \\ K(\bar{x}_{k+N_p,k}^* + \sigma_{k+N_p}) \end{bmatrix}_{pN_p \times 1} \quad (50)$$

where $CBK\sigma_{k+N_p} = -w_{k+N_p}$. By assumption, and because $\bar{x}_{k+N_p,k} \in \Theta \dot{-} (-CBK) \odot \Theta_w$, such that $\bar{x}_{k+N_p,k}^* + \sigma_{k+N_p} \in \Theta \dot{-} (-CBK) \odot \Theta_w \oplus (-CBK) \odot \Theta_w = \Theta$. According to Lemma 1, it is clear that $u_{\min} < K(\bar{x}_{k+N_p,k}^* + \sigma_{k+N_p}) < u_{\max}$. Therefore, the chosen control sequence satisfies constraint (47).

At this point, the state sequence and output sequence corresponding to the preselected control sequence are as follows:

$$\begin{aligned} & \bar{x}_{k+1+i,k+1} = \\ & \begin{cases} \bar{x}_{k+1+i,k}^*, i = 0, 1, \dots, N_p - 1 \\ (A+BK)\bar{x}_{k+1+i,k}^* + BK\sigma_{k+N_p} - B_w \hat{w}_{k+N_p}, i = N_p \end{cases} \end{aligned} \quad (51)$$

$$\bar{y}_{k+1+i,k+1} = \begin{cases} \bar{y}_{k+1+i,k}^*, i = 0, 1, \dots, N_p - 2 \\ C \bar{x}_{k+1+i,k}^*, i = N_p - 1 \end{cases} \quad (52)$$

Subsequently, the cost function can be formulated for the $k + 1$ th moment:

$$\begin{aligned} J_{k+1} &= \frac{1}{2} \left[\sum_{i=0}^{N_p-1} (\| \bar{y}_{k+1+i,k+1}^* \|_Q + \| u_{k+1+i,k+1}^* \|_R) \right] + \frac{1}{2} \| \bar{x}_{k+1+N_p}^* \|_P \\ &= \frac{1}{2} \left[\sum_{i=0}^{N_p-2} (\| \bar{y}_{k+1+i,k+1}^* \|_Q + \| u_{k+1+i,k+1}^* \|_R) \right] \\ &\quad + \frac{1}{2} \left[\| \bar{x}_{k+N_p,k}^* \|_{C^T Q C} + \| \bar{x}_{k+N_p,k}^* \|_{K^T R K} \right. \\ &\quad + \| \bar{x}_{k+N_p,k}^* \|_{(A+BK)^T P (A+BK)} \\ &\quad \left. + \| K \sigma_{k+N_p} + B_w \hat{w}_{k+N_p} \|_P \right] \end{aligned}$$

where, $\|x\|_* = x^T * x$, and so forth.

And furthermore, we have:

$$\begin{aligned} J_{k+1} &= \frac{1}{2} \left[\sum_{i=1}^{N_p-1} (\| \bar{y}_{k+i,k}^* \|_Q + \| u_{k+i,k}^* \|_R) \right] \\ &\quad + \frac{1}{2} \| \bar{x}_{k+N_p,k}^* \|_{(C^T Q C + K^T R K + (A+B_u K)^T P (A+B_u K))} \\ &= \frac{1}{2} \left[\sum_{i=0}^{N_p-1} (\| \bar{y}_{k+i,k}^* \|_Q + \| u_{k+i,k}^* \|_R) \right] + \frac{1}{2} \| \bar{x}_{k+N_p,k}^* \|_P \\ &\quad - \frac{1}{2} (\| \bar{y}_{k,k}^* \|_Q + \| u_{k,k}^* \|_R) \\ &= J_k^* - \frac{1}{2} (\bar{y}_{k,k}^{*T} Q \bar{y}_{k,k}^* + u_{k,k}^{*T} R u_{k,k}^*) \\ &= J_k^* - \frac{1}{2} (\bar{x}_k^T C^T Q C \bar{x}_k + u_k^T R u_k) \end{aligned}$$

At this point, if there is an optimized solution at moment $k + 1$, then there must be:

$$J_{k+1}^* \leq J_{k+1} \leq J_k^* - \frac{1}{2} x_k^T C^T Q C x_k \quad (53)$$

J_k satisfies the discrete Lyapunov condition, Lemma 2, and thus the nominal system is asymptotically stabilized under this control strategy.

3.4.2. Stability of primary system

The error equation $e_{k+1} = A e_k + \delta w$ is the uncontrollable part of the control strategy presented in Section 3.2. It is easy to obtain the error at the N th moment as:

$$e_N := A^N e_0 + \sum_{i=0}^{N-1} A^i \delta w_i \quad (54)$$

The model prediction starts by considering $e_0 = 0$, defining the set of errors as:

$$S(N) := \sum_{i=0}^{N-1} A^i \delta w = A^0 \delta w \oplus A^1 \delta w \oplus \dots \oplus A^{N-1} \delta w \quad (55)$$

Then, an additional constraint on the feasible domain for problem (48) is as follows:

$$\begin{aligned} &\min_{U_k} J, \\ J &= \frac{1}{2} \left[\sum_{i=0}^{N_p-1} (y_{k+i,k}^T Q y_{k+i,k} + u_{k+i,k}^T R u_{k+i,k}) \right] + \frac{1}{2} x_{k+N_p}^T P x_{k+N_p} \\ &\text{s.t.} \begin{cases} u_{\min} \leq u_{k+i,k} \leq u_{\max}, i = 0, 1, \dots, N_p - 1 \\ y_{\min} \leq y_{k+i,k} \leq y_{\max}, i = 1, 2, \dots, N_p \\ x_{k+i,k} \in \Theta_1, i = 0, 1, \dots, N_p - 1 \\ x_{k+N_p,k} \in \Theta_1 \ominus (-CBK) \odot \Theta_w \end{cases} \end{aligned} \quad (56)$$

where $\Theta_1 = \Theta \ominus S(N_p)$ and for operator $\odot$, $x \in \Theta_1$.

Based on the nominal system calculation, when the system's state quantity $x \in \Theta_1$, accounting for the error system, its state value will assuredly remain within the stability domain Θ within N_p steps, i.e. $x \in \Theta_1 \oplus S(N_p) = \Theta$. The system will then consistently operate within

Table 1

MPC simulation verification parameters.

Item	Value
Sample time	0.1 s
Controller window size N_p	20
Expanding ratio k_{ratio}	1×10^9
Input weights R	$diag(1, 1, 1) \times 10^{-2}$
Output error weights Q	$diag(1, 1, 1)$
Terminal output error weights S	$diag(1, 1, 1)$
Input upper limit	$[0.1 \ 2 \ 4]^T$
Input lower limit	$-[0.1 \ 2 \ 4]^T$
Input rate of change upper limit (/s)	$[0.05 \ 0.5 \ 1]^T$
Input rate of change lower limit (/s)	$-[0.05 \ 0.5 \ 1]^T$

a pipeline of state quantities, ensuring it remains within bounds Θ . Furthermore, according to Theorem 2, the system maintains stability.

This suggests that if the predictive ability of the LSTM is poor, it narrows the feasible domain of the controller, making the control strategy conservative, and vice versa, it increases the degree of freedom of the control strategy and improves the control effect.

To guarantee the stability of the initial control strategy, we must first identify the maximum set of states Θ , the set of LSTM forecast values Θ_w , and the set of forecast errors $\Theta_{\delta w}$ that comply with the controller constraints and stabilize the system. Next, we calculate the set of errors $S(N_p)$ and subsequently adjust the state constraints and terminal constraints. This process ensures the system's stability under this control strategy.

4. Simulation verification

In this paper, based on the PM wave spectrum, long crest waves are considered, and a significant wave height of 8 m, wave frequency number of 50, and speed of 5.1 m/s are selected for simulation verification.

It must be taken into account that the actual USV sailing will encounter the limitations of arithmetic power, the errors of sensors, and the limitations of actuators. For one thing, the MPC is an optimization class controller, which has a high demand for arithmetic power, so the control period cannot be too small; for another, a series of problems such as sensor errors, actuator calibration errors, system delays and so on may occur, and the immunity to disturbances of the control algorithms must be verified; finally, the actuator has certain limitations on the inputs to the model, which are mainly reflected in the amplitude and rate of change limitations of the control quantities.

Simulation experiments have found that the MPC-type active decoupling control strategy is most significantly affected by the model state measurement error and the rate of change of the control quantity inputs, therefore, this chapter will gradually add non-ideal factors to the model in order to test the control performance of the MPC+LSTM control strategy proposed in this paper.

Meanwhile, simulation experiments have revealed that significant disparities between input and output magnitudes pose challenges in determining appropriate weight coefficients. Additionally, a broad solution domain for the solver impedes iteration speed. Thus, a proposed solution involves expanding the original model's B matrix by a factor of k_{ratio} , and subsequently amplifying the signal at the controller's output port by the same factor, aiming to enhance the performance of the MPC controller.

In this paper, the simulation verification parameters are selected as follow in Tables 1 and 2:

Below, the MPC+LSTM attitude control strategy proposed in this paper is simulated and validated across four distinct scenarios. Attitude control effectiveness is compared under identical constraints for PID, LQR, MPC, feed-forward perturbation compensation MPC (MPC+FC1) based on the nominal system free of disturbances (Liang and Wen, 2017), feedback compensation MPC (MPC+FC2) based on the perturbed system, and the novel control strategy (MPC+LSTM) introduced

Table 2
LSTM simulation verification parameters.

Item	Value
Training solver	Adams
Number of iterations	160
Initial learning rate	0.01
Learning rate drop period	100
Rate of decline in learning rate	0.15
Number of LSTM cells	200
Number of input layer	3
Number of fully connected layer	3

Table 3
Control parameters when using PID control.

	Kp($\times 10^8$)	Ki ($\times 10^8$)	Kd ($\times 10^8$)
Roll	1	0.5	2
Pitch	0.1	0.0001	50
Yaw	10	2	30

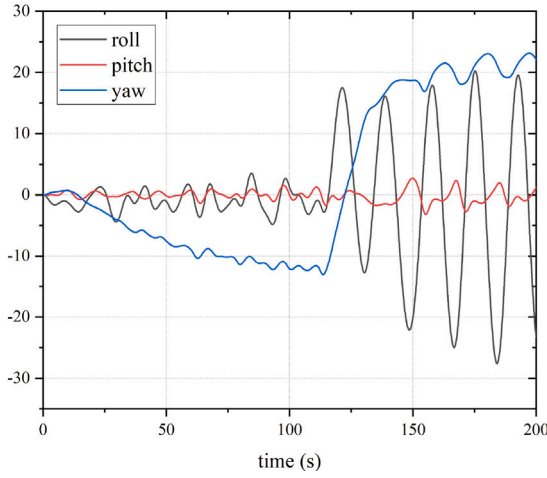


Fig. 7. Roll, pitch and yaw time curves of a USV in the absence of control.

herein. Notably, PID controller parameters can be transformed into a transfer function matrix via the state space equation, and subsequently adjusted in the frequency domain. Tuning outcomes are presented in Table 3.

For the remaining controllers, the weighting factors were kept consistent with Table 1.

4.1. Scenario I: no input rate of change limit

The input rate of change limitations outlined in Table 1 are initially disregarded. In other words, the controller actuator is assumed to possess a high input rate of change and is fully equipped to accommodate fluctuations in external wave disturbances.

4.1.1. Without state feedback noise

When there is no control, the duration curves of the USV's roll, pitch, and yaw are depicted in Fig. 7. The figure illustrates that due to the transverse stabilizing center height being significantly smaller than the longitudinal stabilizing center height, the roll amplitude is notably larger than the pitch amplitude. In rough sea conditions, the USV's normal operations could be greatly compromised without attitude control. However, after implementing MPC+LSTM wave compensation to mitigate roll and pitch amplitudes, the vehicle's stability is enhanced, as demonstrated in Fig. 8.

It can be seen that the roll and pitch reduction effect of MPC+LSTM is significant and almost realizes complete attitude stabilization.

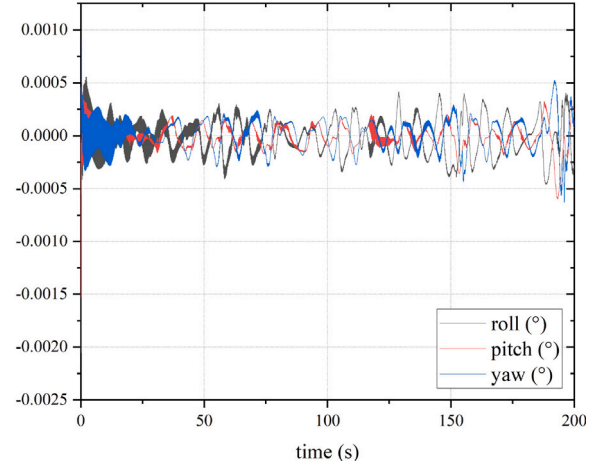


Fig. 8. Roll, pitch and yaw angle ephemeral curves (heading angle 0°) of the navigator using MPC+LSTM wave compensation.

Table 4
Comparison of standard deviation of attitude angle under various control strategies.

	Roll (°)	Pitch (°)	Yaw (°)
Original SD	9.4611	0.9426	13.1039
PID	0.06369	0.14836	0.17543
LQR	0.0022	0.00542	0.0088
MPC	0.00194	0.0012	0.00204
MPC+FC1	2.6983×10^{-4}	1.6956×10^{-4}	2.4334×10^{-4}
MPC+FC2	2.6882×10^{-4}	1.3005×10^{-4}	2.4350×10^{-4}
MPC+LSTM	1.7939×10^{-4}	1.1843×10^{-4}	1.4677×10^{-4}

In the case of no input rate of change limitation and no state feedback noise, compared with the control strategies proposed by past scholars (Jimoh et al., 2021; Liang and Wen, 2017), the scheme proposed in this paper has certain advantages, mainly reflected in the “peak shaving” of the attitude angle time curve. As shown in Fig. 9.

To quantitatively compare the control effects of various strategies, we propose using the SD (standard deviation) value of each attitude angle as an index for evaluation, which can be calculated as follow:

$$I_{SD} = \sqrt{\sum_{i=1}^n \zeta_i^2}, \zeta \in \{\phi, \theta, \psi\} \quad (57)$$

Calculating the standard deviation of the attitude angles before and after the control is applied by the USV, we get Table 4.

Utilizing the data from the time-attitude curves, the boxplot for each control strategy is depicted, as illustrated in Fig. 10.

The optimization-class control strategy demonstrates significant superiority over PID, especially when input rate of change constraints and state feedback noise are absent. Even compared to the least effective LQR strategy, it results in a decrease of 96.55%, 96.35%, and 94.83% in the standard deviation of attitude angle, respectively, compared to PID. Compared to the conventional MPC+FC1 and MPC+FC2 controllers, the average decrease is 34.45% and 27.31%, respectively (average of the decrease in standard deviation of the three degrees of freedom). In this scenario, the performance of MPC and LQR appears quite similar. The three control strategies compared in Fig. 9, on the other hand, outperform the MPC control and MPC+LSTM>MPC+FC2>MPC+FC1.

4.1.2. With state feedback noise

In the real-world operation of the USV, various issues such as sensor errors, actuator calibration discrepancies, system delays, and so forth may arise. Therefore, it becomes imperative to assess the

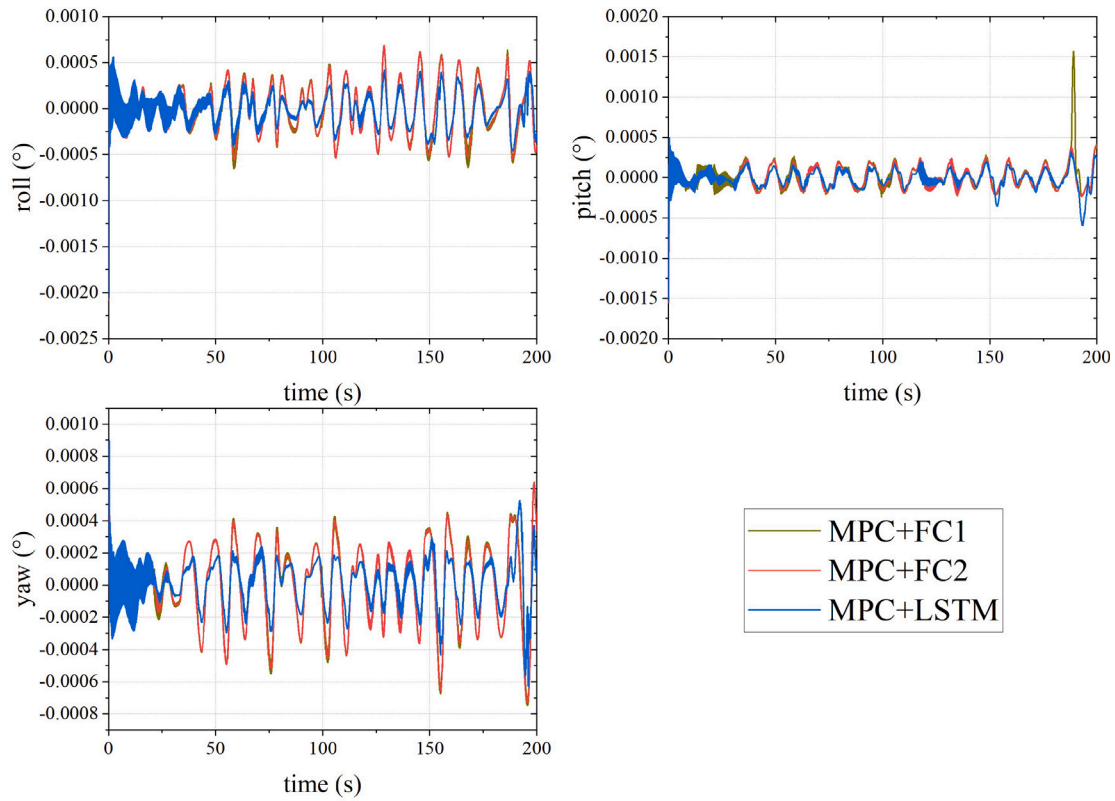


Fig. 9. Comparison of disturbance compensation strategies without input rate of change limitation and state feedback noise.

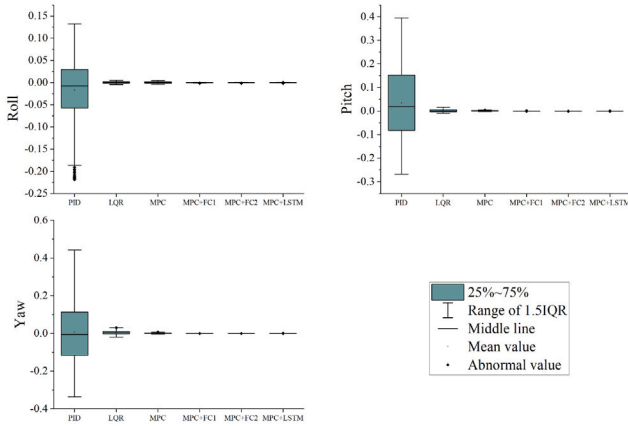


Fig. 10. Boxplot of each control strategy (without state feedback noise).

disturbance rejection capabilities of several control strategies. With the control parameters of each controller held constant, model state feedback is introduced as a normally distributed disturbance signal with a mean value of 0 (sampling time of 0.1 s). The standard deviation of disturbance Δx for each state is then set as follows:

$$\Delta x = \begin{bmatrix} 0.0017 & 0.0017 & 0.0017 & 0.0017 & 0.0017 \\ 0.0017 & 0.1 & 0.1 & 0.1 \end{bmatrix}^T$$

The final effect of each control strategy is shown in Table 5 and Fig. 11.

Analysis of Table 5 and Fig. 11 reveals that PID performance exhibits good stability in the presence of state feedback errors. However, the MPC+LSTM control strategy proposed in this study consistently outperforms the PID controller across various metrics. This indicates the robustness of control strategies relying on state feedback, such as MPC

Table 5

Comparison of standard deviation of attitude angle under various control strategies (with state feedback noise).

	Roll (°)	Pitch (°)	Yaw (°)
PID	0.11379	0.15729	0.19164
LQR	0.22016	0.0884	0.07729
MPC	0.1059	0.09247	0.11507
MPC+FC1	0.13557	0.12201	0.14553
MPC+FC2	0.28115	0.2218	0.26158
MPC+LSTM	0.10824	0.09582	0.11008

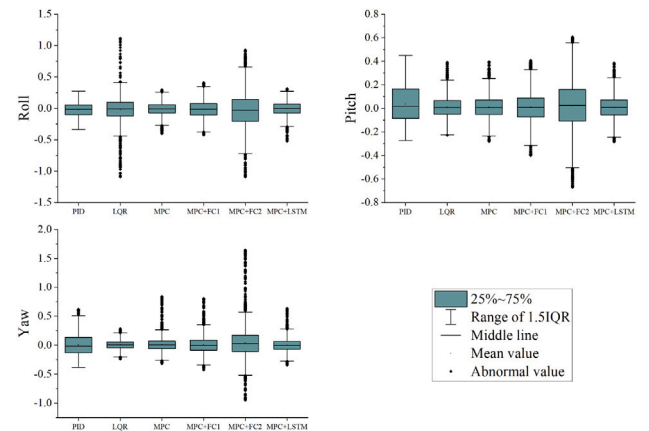


Fig. 11. Boxplot diagrams for each control strategy (with state feedback noise).

and LQR, particularly in handling deviations in state variables. Conversely, the MPC+FC2 control strategy amplifies measurement noise,

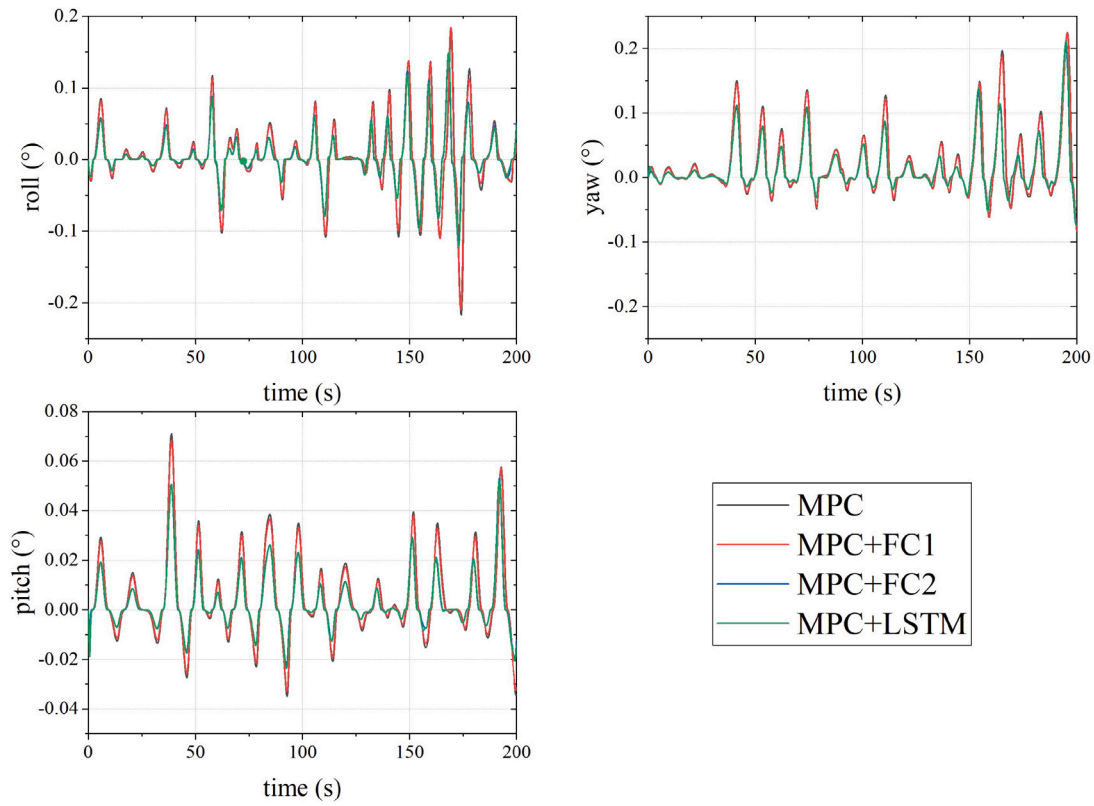


Fig. 12. Comparison of the attitude control effect of each control strategy with input rate of change limitation and without state feedback noise (MPC class controller).

Table 6

Comparison of standard deviation of attitude angle under various control strategies with input rate of change limitation and without state feedback noise.

	Roll (°)	Pitch (°)	Yaw (°)
PID	0.06706	0.1483	0.17825
LQR	41.97896	10.28745	41.99386
MPC	0.04745	0.01622	0.0485
MPC+FC1	0.04562	0.01535	0.04699
MPC+FC2	0.03255	0.01101	0.03652
MPC+LSTM	0.0323	0.0111	0.03732

rendering it the least effective among the controllers examined. However, the MPC+LSTM controller effectively captures disturbance trends, mitigating the impact of sensor measurement noise while maintaining satisfactory control performance.

4.2. Scenario 2: with input rate of change limitations

After a series of simulation experiments, this subsection selects the more stringent limits on the input rate of change, which are shown in Table 1.

4.2.1. Without state feedback noise

After having input rate of change limits, the simulation results are shown in Figs. 12 and 13, and Table 6.

At this critical input rate of change limit, both MPC and MPC+FC1 perform similarly, while MPC+FC2 and MPC+LSTM also exhibit similar performance. Interestingly, the simulation results indicate that the LQR controller not only fails to stabilize the attitude, but also exacerbates the oscillation of the USV under more severe input variability limits. Conversely, the effect of several disturbance-compensated MPC controls remain good, with $\text{MPC+LSTM} \approx \text{MPC+FC2} > \text{MPC+FC1} > \text{MPC}$. Compared with the PID controller, MPC+LSTM optimized 51.83%, 92.52%, and 79.06% for roll, pitch and yaw respectively

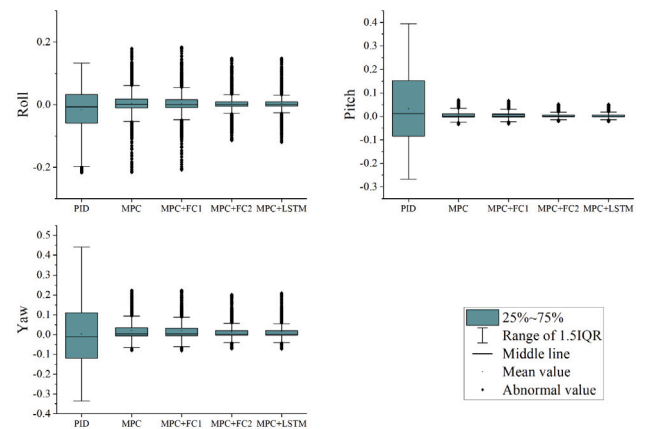


Fig. 13. Boxplot of each control strategy with input rate of change limitation and no state feedback noise (except LQR, no sensor noise).

Table 7

Comparison of standard deviation of attitude angle under various control strategies with input rate of change limitation and state feedback noise.

	Roll (°)	Pitch (°)	Yaw (°)
PID	0.22194	0.17185	1.12825
LQR	40.06138	9.83685	38.91292
MPC	0.21309	0.1908	0.29405
MPC+FC1	0.4091	0.25424	0.24188
MPC+FC2	18.32506	1.95065	5.08968
MPC+LSTM	0.20987	0.20923	0.3127

4.2.2. With state feedback noise

The simulation results after having input rate of change limitations are shown in Fig. 14 and Table 7.

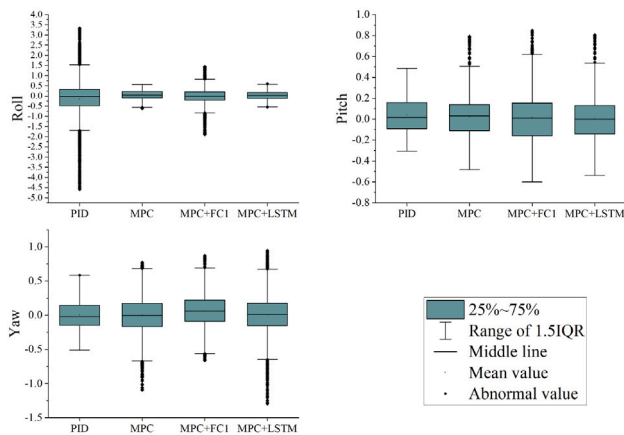


Fig. 14. Boxplot of each control strategy (except LQR, MPC+FC2) with input rate of change limitation, state feedback noise.

The results show that with input change rate limitation and with state feedback noise, the PID is also difficult to maintain a better control effect, and its peak roll angle reaches about 2° . While LQR, MPC+FC2 are all divergent and do not achieve the purpose of stabilizing the USV attitude. Conversely, the MPC and MPC+LSTM control strategies continue to demonstrate superior control effectiveness, with both exhibiting similar performance.

4.3. Wrap-up

Based on the discussion of the above four scenarios, appropriately generalized, we have:

- It is difficult for PID-type control strategy to achieve very good attitude control effect like MPC and LQR-type control strategy, but it is relatively less affected by state quantity feedback noise.
- The disturbance compensation type MPC control strategies (MPC+FC1, MPC+FC2, MPC+LSTM) can further improve the control effect of MPC, but with the enhancement of the input variation rate limitation, the control effect of all the methods is gradually approaching to the MPC control, and in terms of the speed of approaching, the $\text{MPC+FC1} > \text{MPC+FC2} \approx \text{MPC+LSTM}$, while the LQR control is difficult to cope with the input rate of change limitation.
- MPC+FC2 is very sensitive to state-volume feedback noise and performs poorly in the presence of large noise. The optimization class control methods that have some resistance to noise are MPC, MPC+FC1, MPC+LSTM, and under the same noise conditions, it is still the MPC+LSTM control strategy proposed in this paper that has the best effect, but with the increase of noise, the advantage is gradually lost, and the effect is basically the same as that of MPC control.

Taken together, the MPC+LSTM control strategy proposed in this paper is in a better state in all aspects except the algorithm is more complicated, and it has the strongest ability to compensate for wave disturbances.

5. Conclusion

In this paper, for the classical problem of attitude stabilization (sway reduction and heading control) of surface vehicles, a model prediction attitude control strategy based on LSTM short-term wave forecast is proposed. Compared to traditional PID and LQR controllers, there is a significant improvement in control performance. Compared with MPC+FC1 and MPC+FC2 controllers, the average optimization

is 34.45% and 27.31%, respectively, under ideal conditions, and the performance is still better than that of MPC+FC1 (average optimization is 12.37% under input constraints and state-feedback noise) under undesirable factors, and the dispersion phenomenon does not occur as in the case of MPC+FC2. The method can also be generalized to scenarios such as AUVs, ROVs, surface ships, and other scenarios that are susceptible to wave interference. The shortcoming of this method is that if the sea state changes drastically (e.g., a meaningful wave height changes from 1 m to 4 m), the prediction effect of the neural network will decrease, and the neural network needs to be re-trained to maintain its control performance. The two-threaded approach of prediction & training will be considered in the future to make this control strategy more adaptable.

CRediT authorship contribution statement

Liang Hong: Writing – original draft, Software, Methodology, Investigation. **Haitao Liu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Quanshun Yang:** Writing – review & editing, Software. **Jiaxuan Yao:** Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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